

Strong Artificial Intelligence

C Reality

Representing the World

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Technical University of Applied Sciences, Würzburg, 27.03.2026

Repetition „Singularity“

- Which technological unique developments („singularities“) massively changed our daily life and work?
- Which probably will lead to the humanization of machines?
- Why is „Work“ as a concept so important?
- What components make up a „Work System“ and how are the components connected?

Repetition „Intelligence“

- Give a proper definition of intelligence.
- Which components does a machine need, to be intelligent and human like?
- What is the only thing humans can do?

Workshop „Reality“

1. What is reality? Define!
2. How do Computer Scientists handle reality?
3. Why does a digital brain not yet exist, even if hundreds of thousands of computer scientists model the world since more than a half of a century?
4. Is reality subjective or objective?
5. Is there one or are there more realities?

C Reality

- | | |
|----------------------|---------------------------------|
| 1. World | Place of Being |
| 2. Entity | Worlds One Thing and Everything |
| 3. Differentiation | Orientation Giver |
| 4. Representation | Persistencer |
| 5. Primary Reality | My Representation |
| 6. n-ary Reality | Others Representation |
| 7. <i>Conclusion</i> | Paradox and Dualism |



1. World

Orientation Giver

1. World

Everything

around me &



1. World

Fundamentals

What is the World?

What surrounds us with
ourselves included.

Are Minds and Thoughts Part
of the World?

Yes.

Is there one world?

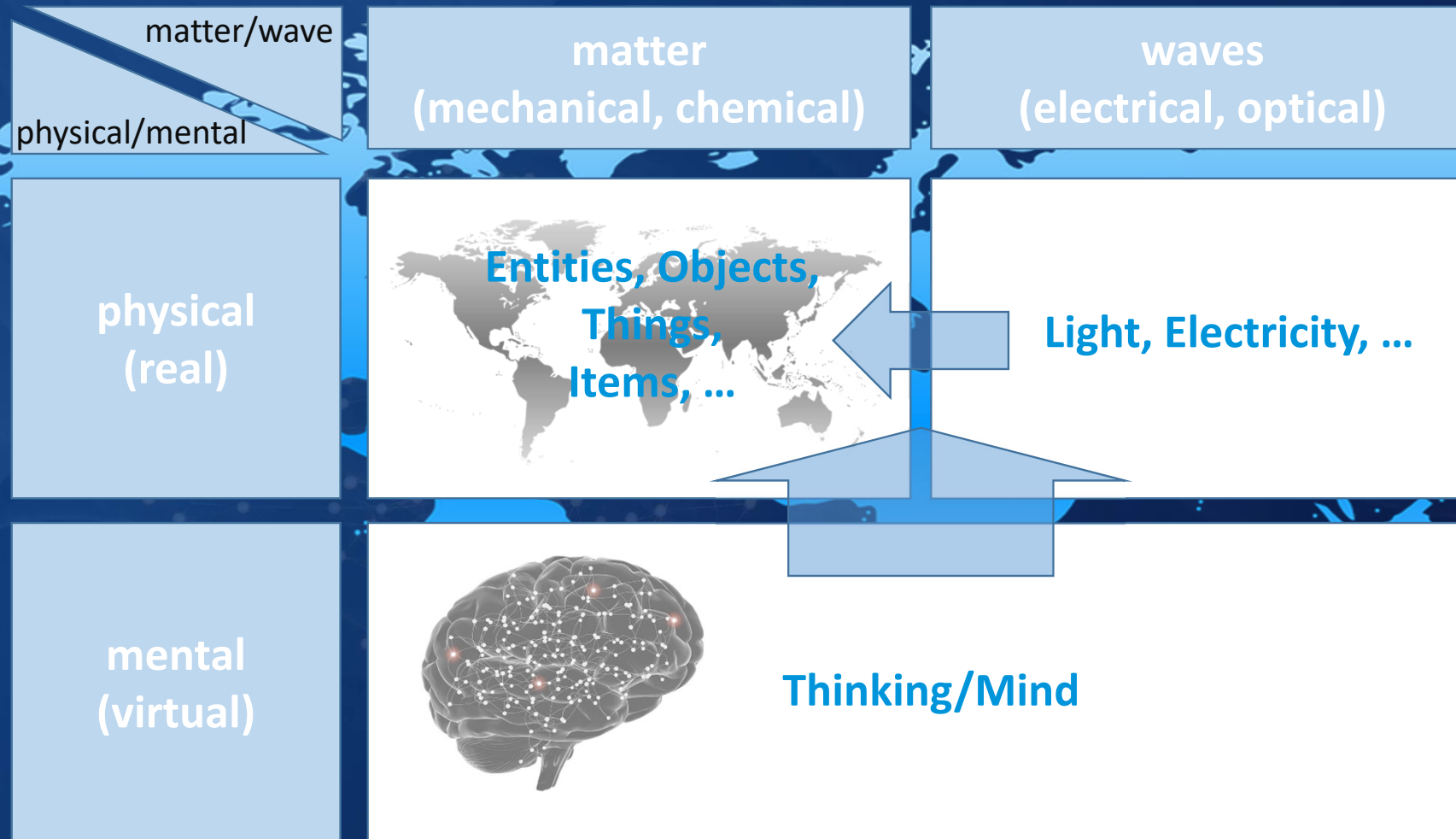
Yes.
We all face the same
one world.

What is not existing?

Virtual things.

1. World

World Phenomenon Matrix



1. World: Change

Fundamentals

What is a Change?

Basically
the movement of something
(change of location).

What Changes?

Objects and Minds.

Who does the Changes?

Nature and Humans.

Are there Other Activities
than Movement?

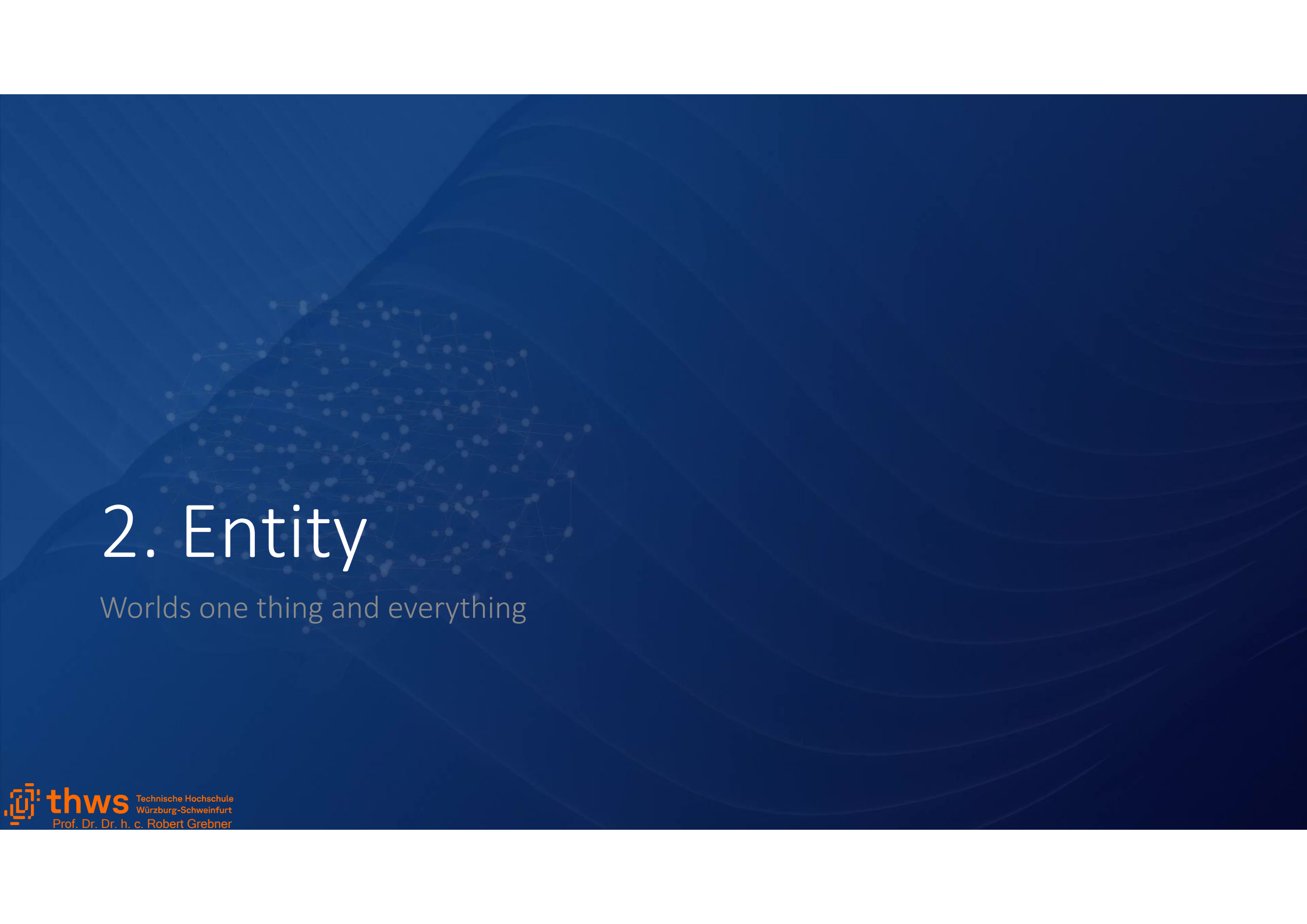
No.

1. World

World Change Matrix

Changer Changed	Nature (natural)	Humans (artificial)	Others (artificial intelligence)
physical	Erosion, Natural Disasters, Plant Life, ...	Any Matter or Wave	Any Matter or Wave
mental	Learn to Speak, Learn to Walk, ... Insights from Observation and Natural Experiments	Mind: Learning from others.	Mind: Learning from others including AI.

- Try to describe a change in the physical world with no movement is happening.
- What moves, when a brain or mind is changed or changes?



2. Entity

Worlds one thing and everything

2. Entity

A dark blue world map with a network of white dots and lines connecting them, representing a global network or data flow. The text "One Thing" is overlaid in large, bold, orange letters.

One Thing

among countless others

2. Entity

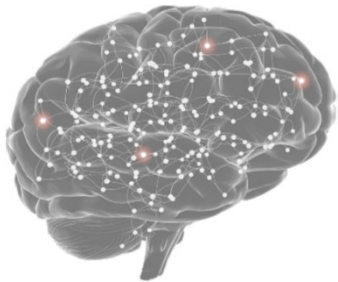
Examples

physical entities (real)



Tree, Person, Laptop, ...

mental entities (virtual)



Thoughts, Action Planing, Verbs, ...

2. Entity

What Characterises an Entity in Principle?

It is delimitable.

What are Delimiters?

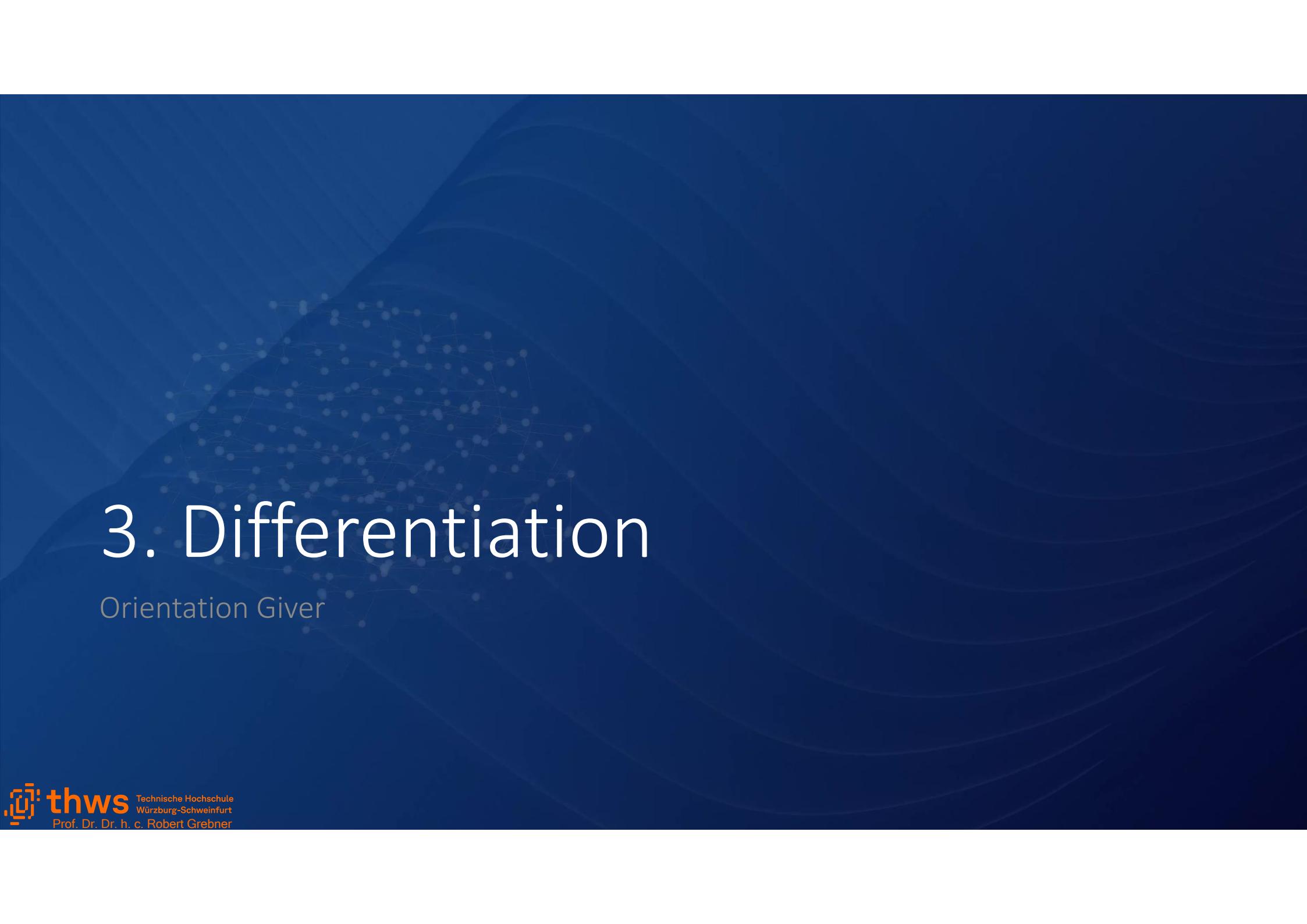
e. g. surfaces, different perceptions, ...

What is a Physical Entity?

A conglomerate of matter and waves.

What is a Mental Entity?

A representation.



3. Differentiation

Orientation Giver

Workshop „Differentiation“

- Develop basic categories for initial distinctions between entities.
- Which categories are most important for us to master our daily life?

3. Differentiation

Categories

?

?

?

?

3. Differentiation

Categories

physical entities

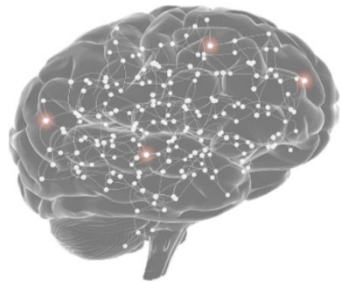


Mobility:
Mobile/Immoble

Intelligence:
Thinker/Non-Thinker

Self-acting:
Dead/Alive

World Substance:
Physical/Mental



Information:
derived by oneself/learned

Information:
verified/unverified

Information:
confidential/non confidential

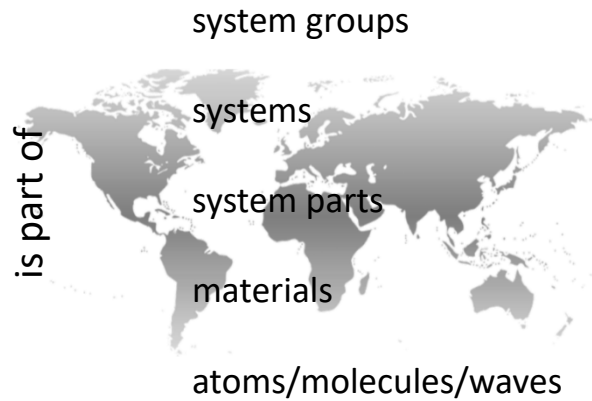
mental entities

3. Differentiation

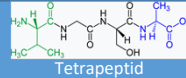
Size Hierarchy: Delimitation via Size Ratio

physical entities

Part Hierarchies
is part of

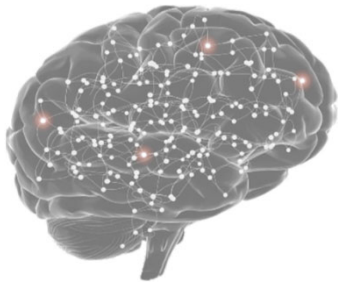


forest	family	computer farm	city
tree	person	computer	house
branch leaf ...	leg arm ...	keyboard ...	brick
biological material		plastic	metal
C ...	Fe Cr ...		Si ...



mental entities

Notion Hierarchies
is a



Representation

Concept

Abstraction

Instantiation

4. Representation

Persistencer

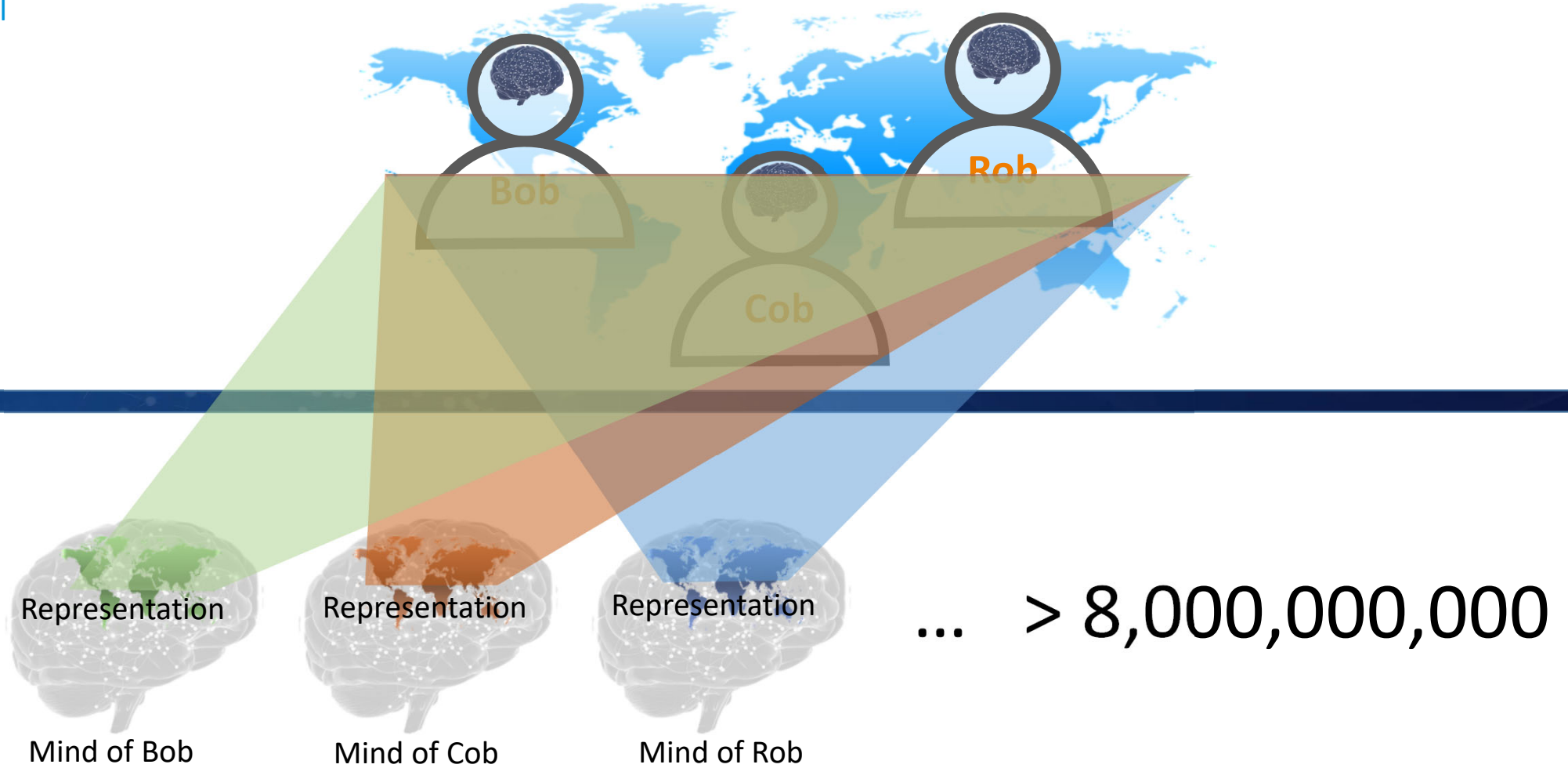
Workshop „Representation“

- What different kind of representations do you know?
- Are representations real or virtual?

4. Representation

physical

mental



4. Representation

Fundamentals

What is a Representation?

A Mental Entity.

Are there Representations in
the Physical World?

No. Except: Artificial Ones
Materialised by Brains.

Can Mental Entities be Anything
Other than Representations?

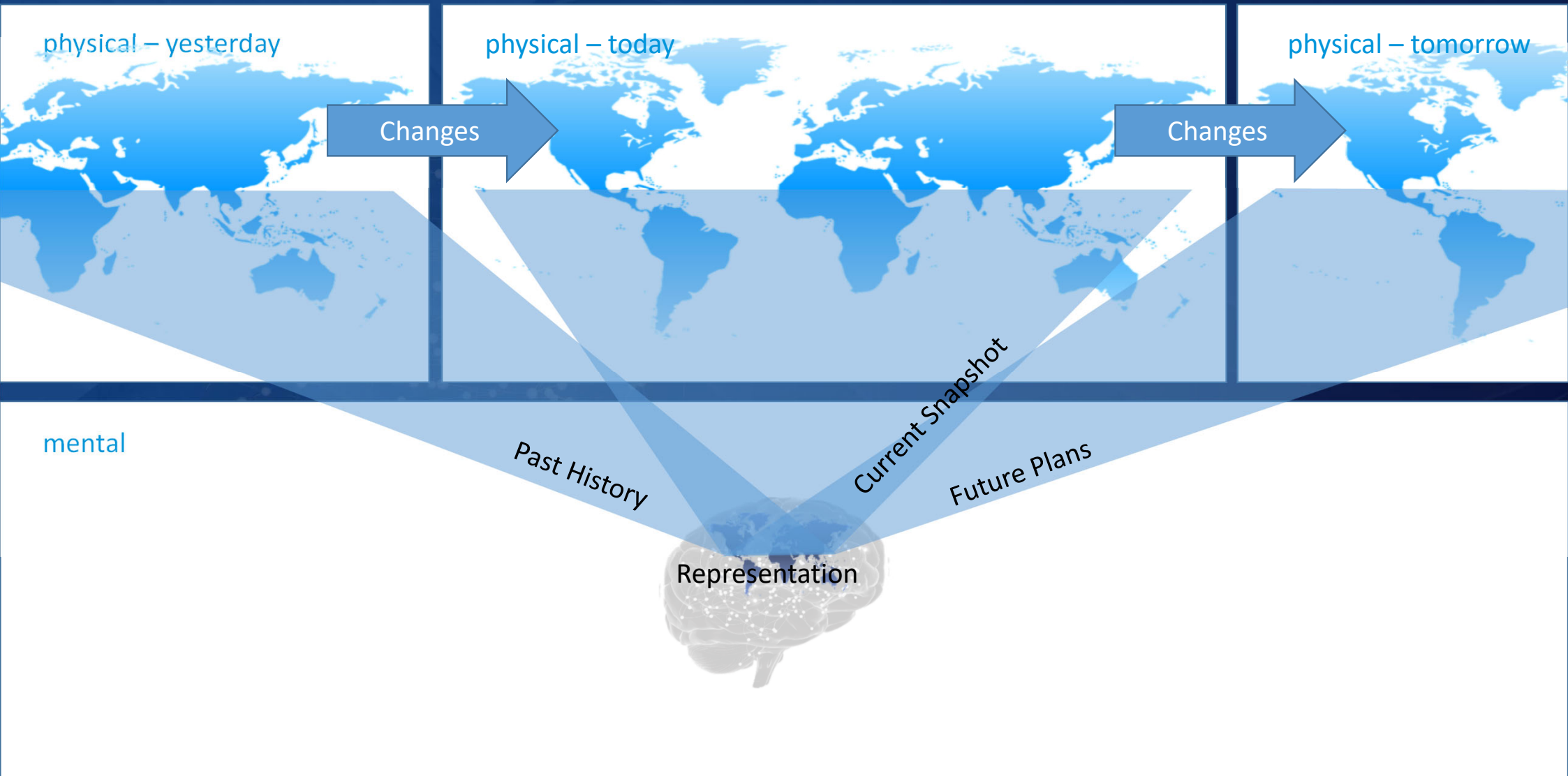
No.

Must Represented Entities
exist?

No.

4. Representation

Time: Past, Present, Future



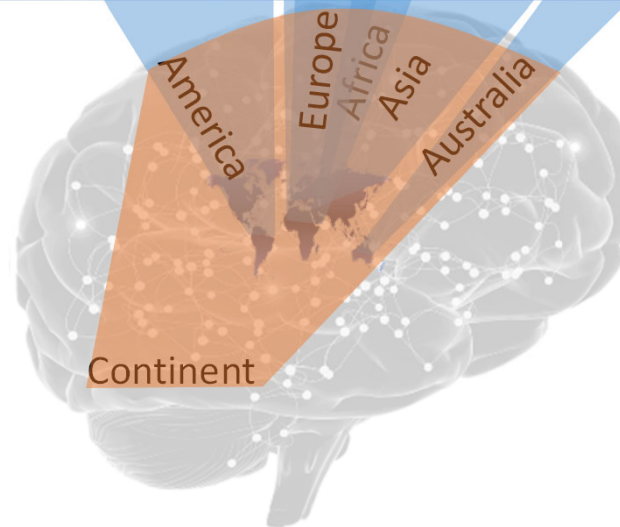
4. Representation

Delimitation: Concrete/Instance, Abstract/Class

physical



mental





5. Primary Reality

My Representation

5. Primary Reality

So What is Reality?

The representation of the world.

How Many Primary Realities Can One Individual have?

One.

What is Primary Reality?

Self experienced representation of the physical world.

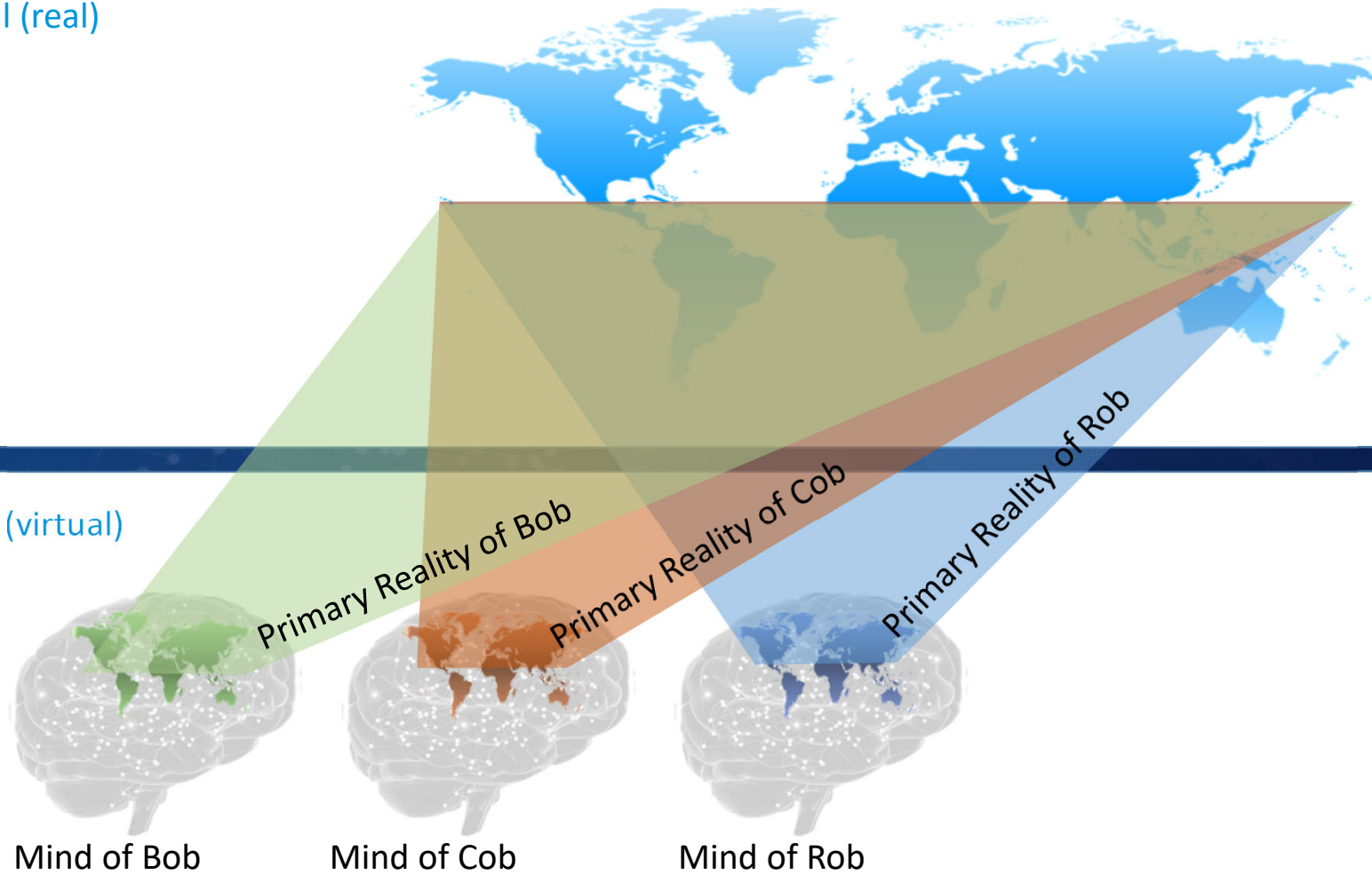
How Many Primary Realities exist?

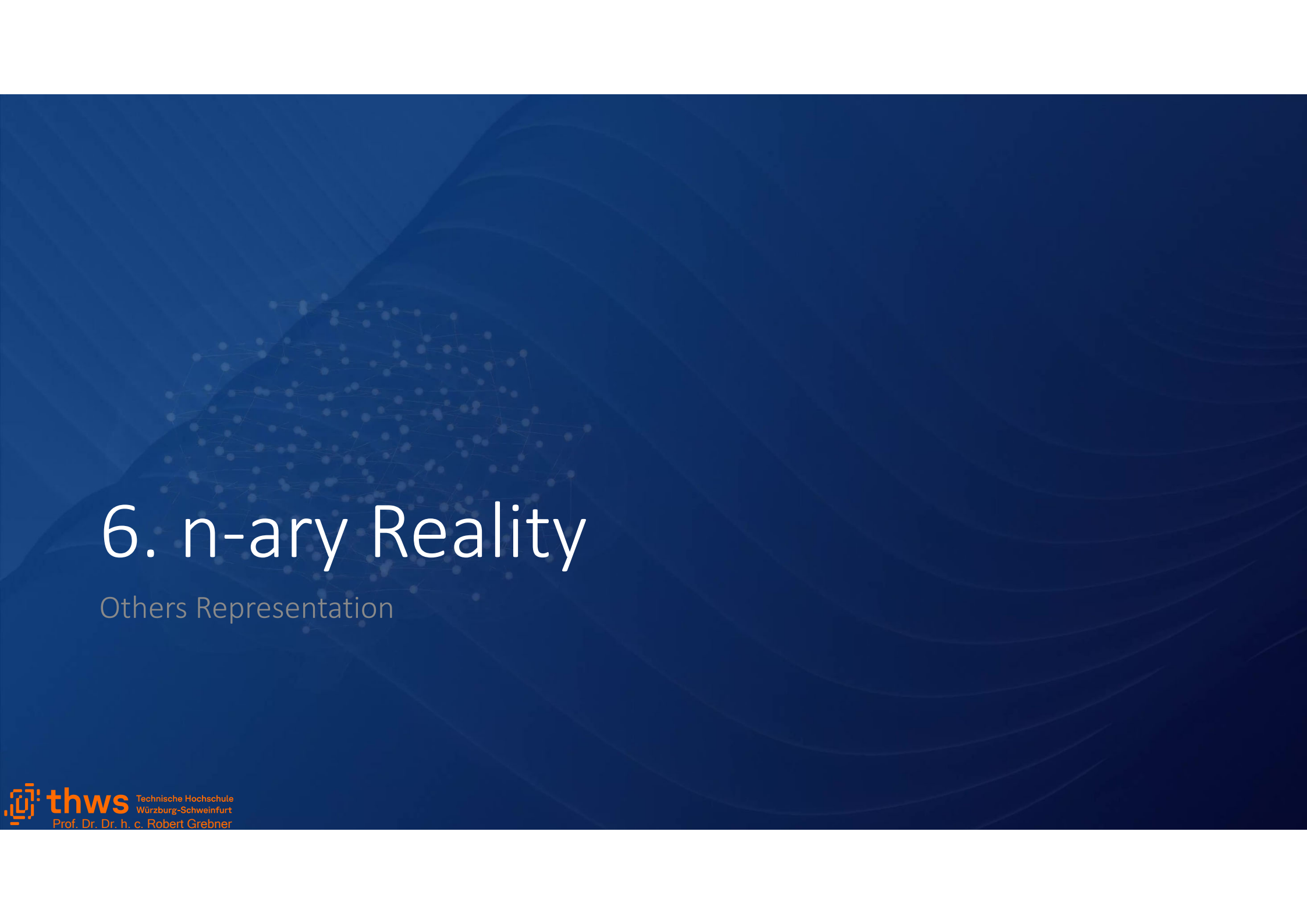
As Much as Individuals.

5. Primary Reality

physical (real)

mental (virtual)





6. n-ary Reality

Others Representation

6. n-ary Reality

What is Secondary Reality?

Representation of the
Primary Reality of Another
Individual.

What is Tertiary Reality?

Representation of a
Secondary Reality of Another
Individual.

What is n-ary Reality?

Representation of a
(n-1)ary Reality of Another
Individual.

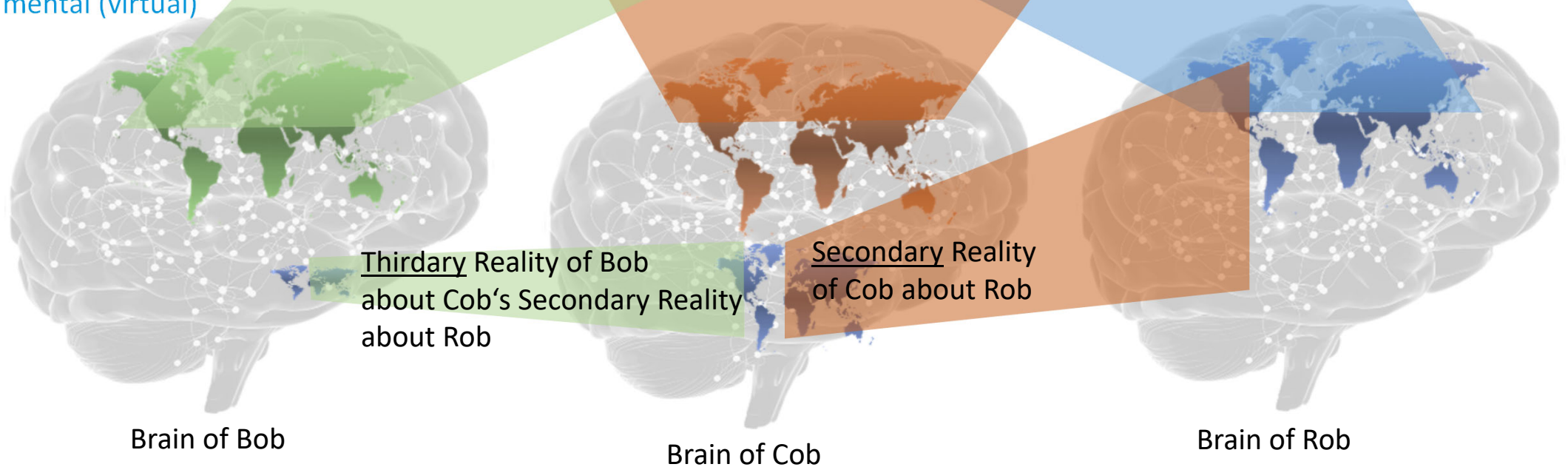
How Many Realities Can One
Individuum have?

Depends on the Amount of
his or her Known Individuals.

6. n-ary Reality

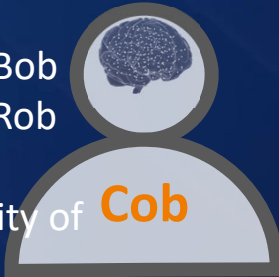
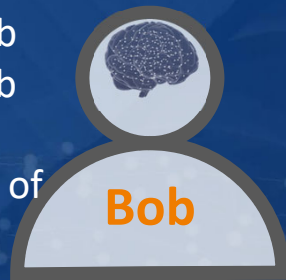
physical (real)

mental (virtual)



6. n-ary Reality

1. Primary Reality
2. Secondary Reality of Cob
3. Secondary Reality of Rob
4. Tertiary Reality of Cob's Secondary Reality of Rob
5. Tertiary Reality of Rob's Secondary Reality of Cob
6. Quarternary Reality of Cob's Tertiary Reality of Rob's Secondary Reality of Bob (myself)
7. Quarternary Reality of Rob's Tertiary Reality of Cob's Secondary Reality of Bob (myself)

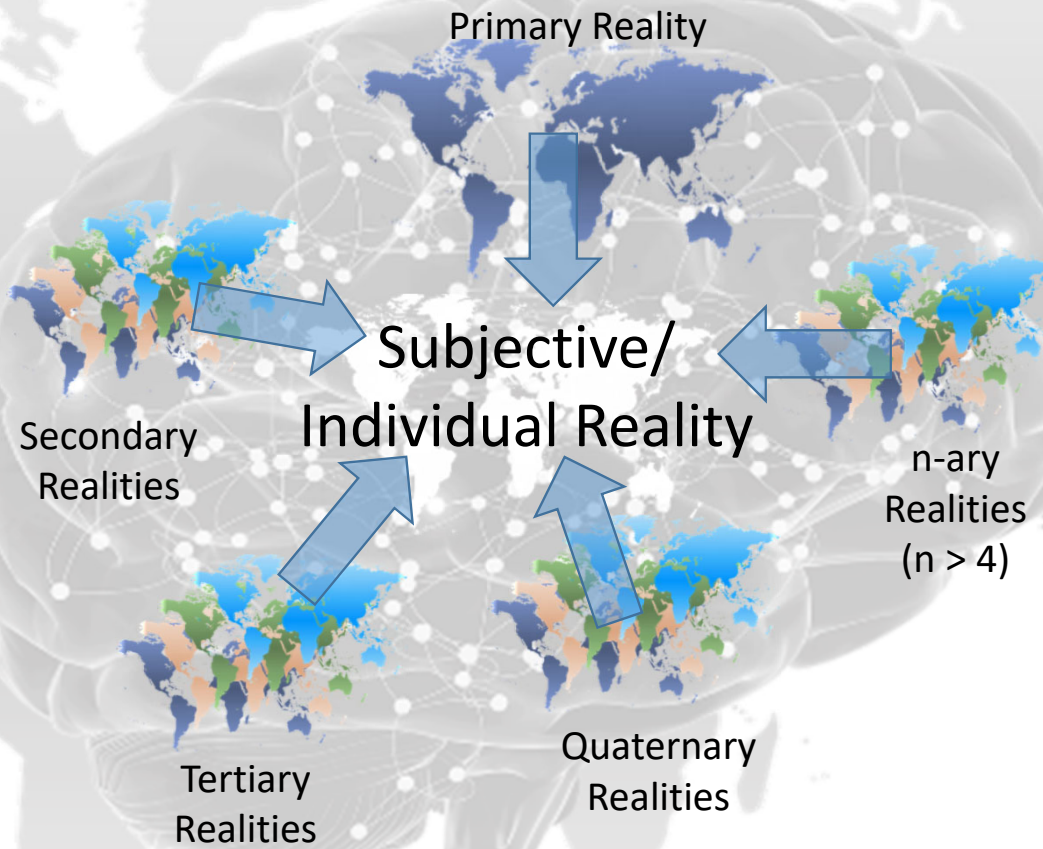


1. Primary Reality
2. Secondary Reality of Bob
3. Secondary Reality of Rob
4. Tertiary Reality of Bob's Secondary Reality of Rob
5. Tertiary Reality of Rob's Secondary Reality of Bob

1. Primary Reality
2. Secondary Reality of Bob
3. Secondary Reality of Cob
4. Tertiary Reality of Bob's Secondary Reality of Cob
5. Tertiary Reality of Cob's Secondary Reality of Bob
6. Quarternary Reality of Bob's Tertiary Reality of Cob's Secondary Reality of Rob (myself)
7. Quarternary Reality of Cob's Tertiary Reality of Bob's Secondary Reality of Rob (myself)
6. Quarternary Reality of Bob's Tertiary Reality of Rob's Secondary Reality of Cob (myself)
7. Quarternary Reality of Rob's Tertiary Reality of Bob's Secondary Reality of Cob (myself)

Objective

Reality



Workshop „Reality description“

Describe a very very small but well known part of the world. We call small parts „discourse world“. The discourse world „K.O.11 today“ shall be our lecture room K.O.11 on 27 March 2026 between 13:30 and now.

- Describe the discourse world in your own words.
- Describe the discourse world using a formal modelling technique or language.

What is the challenge of modelling discourse worlds?



7. Conclusion

Paradox and Dualism

7. Conclusion

Paradox?

- a) Mental entities are part of the mental world.
- b) The mental world takes place in brains.
- c) Brains are part of the real world.
- d) ➔ Mental entities are part of the physical world.

Dualisms?

Thoughts are
PHYSICAL and MENTAL.

Our minds belong to
ourselves AND the world.



I
Thank
You